

A Proof and interpretation of the lambda rule

We use the representation theorem's geometry: $h = 2/N$, $x_j = -1 + jh$, $\gamma = \lambda/h$, and $x_0 = -1$, with $0 < \lambda \leq 1$ and $R = \lceil a_H \sqrt{N} \rceil$ for a fixed halo-design constant $a_H > 0$. The number of neurons is $W = N + 2R + 1$; frequency in grid coordinates is $\theta = h\omega$, so its exact conversion uses N , not W . Our Fourier convention is $\widehat{K}(\xi) = \int_{\mathbb{R}} K(t) e^{-i\xi t} dt$. For $\tanh$, take $K = \text{sech}^2$; then

$$\widehat{K}(\xi) = \frac{\pi\xi}{\sinh(\pi\xi/2)}, \quad \widehat{K}(0) = 2. \quad (\text{A.1})$$

Multiplying K by a constant leaves every ratio below unchanged.

The replica error considered here is the pole contribution $E_{\text{alias}} = \sup_{x \in [-1, 1]} |\mathcal{P}_h[F_x]|$ in the representation proof's signed quadrature identity, with

$$F_x(z) = \frac{a_\gamma(z)}{2} [\tanh(\gamma(x - z)) - \tanh(\gamma(x_0 - z))], \quad a_\gamma(z) = \frac{f(z + id) - f(z - id)}{2id}, \quad d = \frac{\pi}{2\gamma}. \quad (\text{A.2})$$

Horizontal-contour and corrected-boundary errors remain the other representation contributions. The lemmas first keep the actual frequencies of a specified target $P(x) = \sum_j b_j e^{i\omega_j x}$. The local analytic target class is treated separately below; it does not require such a representation.

Lemma A.1 (The repeated spectrum and exact central amplitude). For a finite kernel sum $q(u) = \sum_k a_k K(\lambda(u - k))$,

$$\widehat{q}(\theta) = \lambda^{-1} \widehat{K}(\theta/\lambda) C(\theta), \quad C(\theta) = \sum_k a_k e^{-ik\theta}, \quad C(\theta + 2\pi m) = C(\theta). \quad (\text{A.3})$$

For the $\tanh$ density in (A.2) and $f(x) = e^{i\omega x}$,

$$a_\gamma(x) = i\omega \frac{\widehat{K}(0)}{\widehat{K}(\omega/\gamma)} e^{i\omega x}. \quad (\text{A.4})$$

Sampling this density on the infinite lattice against $K_\gamma(t) = \gamma K(\gamma t)/2$ reproduces the retained derivative amplitude exactly. Integrating gives the output replica multipliers

$$t_m(\lambda, \theta) = \frac{\theta}{\theta + 2\pi m} \frac{\widehat{K}((\theta + 2\pi m)/\lambda)}{\widehat{K}(\theta/\lambda)}, \quad \theta = h\omega, \quad m \neq 0. \quad (\text{A.5})$$

Proof. A change of variables in each kernel transform gives (A.3); integer centers give the periodicity. The shifted values of a tone give $a_\gamma(x) = i \sinh(\omega d) e^{i\omega x}/d$, which is (A.4). To compute the lattice response, the one-periodic function $e^{-i\theta u} \sum_k e^{ik\theta} K(\lambda(u - k))$ has Fourier coefficient $\lambda^{-1} \widehat{K}((\theta + 2\pi m)/\lambda)$ at index m . Indeed, shifting each unit integration interval tiles the real line. Exponential localization justifies this calculation and the convergent series. The density factor in (A.4) cancels the central kernel multiplier. Integration divides the replica at $\omega + 2\pi m/h$ by its frequency, proving (A.5).

More generally, for $K = \psi^{(r)}$, matching the retained amplitude gives $t_m = (\theta/(\theta + 2\pi m))^r \widehat{K}((\theta + 2\pi m)/\lambda)/\widehat{K}(\theta/\lambda)$. This general identity describes the differentiated lattice and its integrated output; the local coefficient formula above is specific to $\tanh$. $\square$

Lemma A.2 (The spectrum bounds the actual pole term). For $0 < |\theta| < \pi$, define

$$S_{K,r}(\lambda, \theta) = \sum_{m \neq 0} \left| \frac{\theta}{\theta + 2\pi m} \right|^r \frac{|\widehat{K}((\theta + 2\pi m)/\lambda)|}{|\widehat{K}(\theta/\lambda)|}. \quad (\text{A.6})$$

For a finite Fourier target with $|h\omega_j| < \pi$, the tanh construction's finite-contour replica contribution obeys

$$E_{\text{alias}}(P) \leq 2 \sum_j |b_j| S_{K,1}(\lambda, |h\omega_j|). \quad (\text{A.7})$$

Constants contribute zero.

Proof. It suffices to consider one unit tone with $\theta > 0$; negative frequencies give the same absolute estimates. Put $s = \pi\theta/(2\lambda)$ and $A = \pi^2/\lambda$. Substituting (A.1) into (A.6) cancels the frequency prefactors:

$$\begin{aligned} S_{K,1}(\lambda, \theta) &= \sinh s \sum_{m \geq 1} [\text{csch}(mA - s) + \text{csch}(mA + s)] \\ &= 4 \sinh s \sum_{\ell \geq 0} \frac{\cosh((2\ell + 1)s)}{e^{(2\ell + 1)A} - 1}. \end{aligned} \quad (\text{A.8})$$

The second line follows by expanding $\text{csch } z = 2 \sum_{\ell \geq 0} e^{-(2\ell + 1)z}$ and summing over m . All terms are positive and the sums converge.

At height $(2\ell + 1)d$, the quadrature integrand has at most four poles, at $x \pm i(2\ell + 1)d$ and $x_0 \pm i(2\ell + 1)d$. Its tanh residues have magnitude $1/(2\gamma)$. By (A.4), the four density magnitudes sum to $4(\sinh s/d) \cosh((2\ell + 1)s)$. The contour formula multiplies residues by 2π and its quadrature factor has magnitude at most $(e^{(2\ell + 1)A} - 1)^{-1}$. Since $\gamma d = \pi/2$, this layer contributes at most

$$8 \sinh s \frac{\cosh((2\ell + 1)s)}{e^{(2\ell + 1)A} - 1}.$$

There are finitely many enclosed layers. Extending this positive scalar sum to all layers and applying (A.8) gives $2S_{K,1}$. Linearity and the triangle inequality give (A.7). No target value outside the local contour is used in this argument. $\square$

The factor two has a concrete origin: the representation is anchored at x_0 , so both x and x_0 contribute residues. Equivalently, anchoring an ideal output subtracts its alias value at x_0 . An unanchored ideal output has the smaller amplitude-sum bound $\sum_j |b_j| S_{K,r}$.

Lemma A.3 (The first pair controls the remaining replicas). Let K have an even, positive Fourier transform, and fix $0 \leq \theta < \pi$. Assume $\log \widehat{K}$ is concave on $[(2\pi - \theta)/\lambda, \infty)$ and

$$\rho = \frac{\widehat{K}((4\pi - \theta)/\lambda)}{\widehat{K}((2\pi - \theta)/\lambda)} < 1. \quad (\text{A.9})$$

Then $2S_{K,r}(\lambda, \theta) \leq \mathcal{R}_{K,r}(\lambda, \theta)$, where

$$\mathcal{R}_{K,r}(\lambda, \theta) = \frac{2 \left[\left(\frac{\theta}{2\pi - \theta} \right)^r \widehat{K}((2\pi - \theta)/\lambda) + \left(\frac{\theta}{2\pi + \theta} \right)^r \widehat{K}((2\pi + \theta)/\lambda) \right]}{(1 - \rho) \widehat{K}(\theta/\lambda)}. \quad (\text{A.10})$$

For $r = 0$, the powers are one; for $r > 0$, set $S_{K,r}(\lambda, 0) = \mathcal{R}_{K,r}(\lambda, 0) = 0$ for constants represented by the output polynomial.

Proof. Concavity makes $\log \widehat{K}(t + 2\pi/\lambda) - \log \widehat{K}(t)$ nonincreasing in t . Every successive transform ratio in either replica sequence is therefore at most ρ . The integration factors also decrease along each sequence. If T_r is the first-pair contribution with its exact denominator $\widehat{K}(\theta/\lambda)$, summing two geometric series gives

$$T_r \leq S_{K,r} \leq \frac{T_r}{1 - \rho}, \quad S_{K,r} - T_r \leq \frac{\rho T_r}{1 - \rho}.$$

The nearer sequence starts at $(2\pi - \theta)/\lambda$ and its next point is $(4\pi - \theta)/\lambda$, giving exactly the stated ρ . The other sequence starts farther out, at $(2\pi + \theta)/\lambda$, so concavity bounds its successive ratios by the

same ρ . For $\tanh$, $\widehat{K}$ decreases and $(\log(t/\sinh t))'' = -1/t^2 + \text{csch}^2 t < 0$ for $t > 0$, since $\sinh t > t$. Thus $0 < \rho < 1$ and the hypotheses hold. $\square$

Combining Lemmas A.2 and A.3 proves the main spectral theorem: $E_{\text{alias}}(P) \leq \sum_j |b_j| \mathcal{R}_{K,1}(\lambda, |h\omega_j|)$. For one absolute frequency this is $S\mathcal{R}_{K,1}(\lambda, h|\omega|)$, where $S = \sum_j |b_j|$. The factor two in (A.10) incorporates anchoring; a first-pair formula for an unanchored output omits it.

The local analytic class and a precision-based rule. The local class $\mathcal{A}_{\delta,B}(I_\Delta)$ consists of real targets holomorphic and bounded by B in the complex δ -neighborhood of $I_\Delta = [-1 - \Delta, 1 + \Delta]$. For such targets, the representation proof directly bounds every density value on the contour by $M_\sigma \leq 4B/\delta$. Its pole estimate gives

$$E_{\text{alias}}(f) \leq \frac{4\pi M_\sigma}{\gamma} \mathcal{U}(A) \leq 4B\mathcal{U}(A), \quad \mathcal{U}(A) = \frac{e^{-A}}{(1 - e^{-A})(1 - e^{-2A})}, \quad A = \frac{\pi^2}{\lambda}. \quad (\text{A.11})$$

The last inequality uses the admissibility condition $\gamma\delta \geq 4\pi$. At $\lambda = 1/4$, $4\mathcal{U}(4\pi^2) = 2.863 \times 10^{-17}$. Thus this choice bounds the replica contribution below binary64 machine epsilon times B for every admissible width and target in the stated class. Resolution, boundary, recovery, and arithmetic contributions still require their own bounds. The coarser representation theorem retains the useful width dependence $128\pi B e^{-\pi^2/\lambda}/(\lambda\delta W)$.

The kernel-only practical rule keeps the same rapidly changing spectral factor,

$$E_{\text{alias}}^{\text{basic}}(\lambda) := \frac{\widehat{K}(2\pi/\lambda)}{\widehat{K}(0)} = \frac{\pi^2/\lambda}{\sinh(\pi^2/\lambda)}. \quad (\text{A.12})$$

This proxy equals 5.651×10^{-16} at $\lambda = 1/4$; setting it equal to 2^{-52} gives $\lambda \simeq 0.24408$. It is not the exact output error at $\theta = 0$: constants are represented exactly. Rather, it retains the exponential scale controlling the nearby replicas while absorbing amplitude and frequency factors into a practical tolerance. The normalized width limit follows directly by continuity: at fixed λ , with $\rho_0 = \widehat{K}(4\pi/\lambda)/\widehat{K}(2\pi/\lambda) < 1$,

$$\lim_{\theta \downarrow 0} \frac{\mathcal{R}_{K,r}(\lambda, \theta)}{4(\theta/(2\pi))^r} = \frac{\widehat{K}(2\pi/\lambda)}{(1 - \rho_0)\widehat{K}(0)}. \quad (\text{A.13})$$

Both first-pair arguments tend to $2\pi/\lambda$, the denominator tends to $\widehat{K}(0)$, and each integration factor divided by $(\theta/(2\pi))^r$ tends to one. When ρ_0 is negligible, the right side is the basic score. This limit holds when the target angular frequency ω is fixed and N grows, since $\theta = 2|\omega|/N$; holding θ fixed instead would preserve the finite-angle ratio. For $r > 0$ the full output bound retains the vanishing frequency factor, rather than converging to the basic score itself. In particular, when θ/λ is small and the first pair dominates, $S_{K,1}(\lambda, \theta) \simeq |\theta|E_{\text{alias}}^{\text{basic}}(\lambda)/\pi$. Since $\theta = h\omega$, this also explains the representation theorem's additional inverse-width factor at fixed target frequency scale.

Why a small frequency prefactor can leave a useful bandwidth default. The normalized limit above does not justify discarding θ^r from an error estimate. It can justify a simpler *bandwidth choice*, because the error changes much faster with λ than an algebraic prefactor changes its threshold. This distinction is quantified as follows. Write $A_K(\lambda) = \widehat{K}(2\pi/\lambda)/\widehat{K}(0)$ only within this derivation, and suppose the small-angle and first-pair reductions are accurate. For the practical first-pair score $T_{K,r}$ used below, they give

$$T_{K,r}(\lambda, \theta) \simeq C_\theta A_K(\lambda), \quad C_\theta = 2(\theta/(2\pi))^r.$$

Let λ_b solve $A_K(\lambda_b) = \varepsilon$, and let λ_s solve $C_\theta A_K(\lambda_s) = \varepsilon$. Both are roots for these stated surrogate scores. The mean value theorem, applied to $t \mapsto \log A_K(e^t)$, gives the exact identity

$$\log \frac{\lambda_s}{\lambda_b} = -\frac{\log C_\theta}{D_K(\tilde{\lambda})}, \quad D_K(\lambda) = \frac{d \log A_K(\lambda)}{d \log \lambda}, \quad (\text{A.14})$$

for some $\tilde{\lambda}$ between the two roots. In particular, if $D_K \geq M > 0$ between them, then $|\log(\lambda_s/\lambda_b)| \leq |\log C_\theta|/M$. This is the precise sense in which the kernel decay dominates a prefactor in the bandwidth calculation.

For tanh, with $a = \pi^2/\lambda$, $A_K = a/\sinh a$ and $D_K = a \coth a - 1$. For exact GELU, $\psi(x) = x\Phi(x)$ with Φ the standard normal CDF, $r = 2$, $K(x) = (2 - x^2)\phi(x)$ with $\phi = \Phi'$ the standard normal density, and $\widehat{K}(\xi) = (1 + \xi^2)e^{-\xi^2/2}$. Writing $z = (2\pi/\lambda)^2$ gives $A_K = (1 + z)e^{-z/2}$ and $D_K = z - 2z/(1 + z)$. At the binary64 basic roots, these logarithmic slopes are approximately 39.44 and 78.92, respectively. A multiplicative prefactor can therefore change the error estimate substantially while moving its bandwidth threshold much less. At high precision, D_K is of order $\log(1/\varepsilon)$, so $|\log C_\theta| \ll \log(1/\varepsilon)$ is a useful regime for the basic approximation, provided the spectral shifts can also be neglected.

The last condition matters: a small grid angle alone does not ensure a small change deep in a Fourier tail. For tanh, putting $s = \pi\theta/(2\lambda)$ and $a = \pi^2/\lambda$, the first pair with its exact denominator, before anchoring, is

$$\sinh s [\operatorname{csch}(a - s) + \operatorname{csch}(a + s)] \simeq 2 \sinh(2s)e^{-a}.$$

The second expression requires $a - s \gg 1$; reducing it further to $2(\theta/(2\pi))A_K(\lambda)$ requires $\pi\theta/\lambda \ll 1$. For GELU, the shifted exponential factors are $e^{-2\pi^2/\lambda^2 \pm 2\pi\theta/\lambda^2}$, so the corresponding reduction requires $2\pi\theta/\lambda^2 \ll 1$ as well as $\theta/\lambda \ll 1$ to replace the central transform by its value at zero. The refined score retains these shifts and the original denominator.

For the plotted sine mixture, the representative frequency is $\bar{\omega} = 30\pi/7$, so the tanh first-pair prefactor is $C_\theta = 60/(7N)$. At $N = 512$, it is about 0.01674: dropping it changes the small-angle score by a factor of about 60. Nevertheless, solving the full first-pair score gives $\lambda = 0.27189$, compared with the basic 0.24408, a change of about 11.4%. This example illustrates threshold sensitivity for the practical score. At $N = 32$ the small-angle reduction is inaccurate, and the full first-pair score instead gives 0.20048.

There is no fixed-precision theorem that the refined bandwidth converges to the basic bandwidth as $N \rightarrow \infty$: for $r > 0$, C_θ vanishes and $|\log C_\theta|$ eventually becomes large. The basic rule is a useful leading-order default in the stated regime, not a uniform limit in width. The independent local-class estimate in (A.11) is what supports the tanh quarter-scale alias guarantee over all admissible widths.

A recipe using ordinary numerical inputs. For an interval of length L divided into N spacings, set $h = L/N$. Once the rule returns λ , use $\gamma = \lambda/h$ in $\psi(\gamma(x - c_j))$. The basic rule needs only the activation and an effective tolerance e_{tol} ; the following equations can be solved by bisection, without a network fit or a numerical Fourier transform:

activation	basic equation	λ (fp32)	λ (fp64)
tanh	$(\pi^2/\lambda)/\sinh(\pi^2/\lambda) = e_{\text{tol}}$	0.50325	0.24408
GELU	$(1 + 4\pi^2/\lambda^2)e^{-2\pi^2/\lambda^2} = e_{\text{tol}}$	1.00257	0.69857

The table chooses machine epsilon as the effective tolerance. It is the spacing above one: 2^{-23} for binary32 and 2^{-52} for binary64. These equations define practical scores; the tanh representation additionally has its admissibility conditions.

The refinement needs just one additional input: a rough angular frequency scale $\bar{\omega}$ for the target. A typical wavelength ℓ gives $\bar{\omega} \approx 2\pi/\ell$; equivalently, k oscillations across an interval of length L give $\bar{\omega} \approx 2\pi k/L$. For a mixture of known components, an amplitude-weighted average is one possible estimate,

not a required definition of $\bar{\omega}$. The displayed examples retain expC08’s estimates:

target	$\bar{\omega}$
$\sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x)$	$30\pi/7$
$\exp(\sin(3\pi x))$	6.34919

The second has Fourier harmonics at $3\pi n$ with coefficient magnitudes $I_{|n|}(1)$, where I_n is the modified Bessel function; the displayed scale is their amplitude-weighted mean absolute frequency, including the constant component in the normalization. A Fourier estimate from samples is another way to select the scale; a discrete Fourier transform assumes a periodic extension, so endpoint handling can affect the estimate. None of these estimates requires supplying a separate amplitude to the practical selector.

Let $T_{K,r}(\lambda, \theta)$ denote the first-pair expression on the left of (3). Lemma A.3 gives $S_{K,r} \leq T_{K,r}/(1 - \rho_K)$ with the exact denominator $\widehat{K}(\theta/\lambda)$. Thus the simplified score uses that exact denominator, omits the negligible later-pair correction, and absorbs constant factors into the effective tolerance. For tanh, the first-pair score can be evaluated directly as

$$T_{K,1}(\lambda, \theta) = \sinh s [\operatorname{csch}(a - s) + \operatorname{csch}(a + s)], \quad s = \frac{\pi\theta}{2\lambda}, \quad a = \frac{\pi^2}{\lambda};$$

for GELU, substitute $r = 2$ and $\widehat{K}(\xi) = (1 + \xi^2)e^{-\xi^2/2}$ into the same first-pair equation. Both scores increase with λ on the plotted interval $[0.03, 1.5]$ for $0 < \theta < \pi$, so a bracketed scalar solve suffices. For tanh, $t \coth t$ increases for $t > 0$, hence each ratio $\widehat{K}(b/\lambda)/\widehat{K}(d/\lambda)$ increases when $b > d > 0$. For GELU, its logarithmic derivative in λ is $g(b/\lambda) - g(d/\lambda)$, where $g(t) = t^2 - 2t^2/(1 + t^2)$. This difference is positive when $b/\lambda > 1$: g is nonpositive on $[0, 1]$ and strictly increasing above one. The first alias satisfies $b \geq 2\pi - \theta > \pi$, which ensures that condition throughout the plotted interval. The standalone selector checks this condition if a different GELU search interval is supplied.

The effective tolerance e_{tol} is chosen by the user; it is not required to equal machine epsilon. Its role is to absorb the target’s amplitude and fixed normalization constants as well as specify the desired numerical scale. In the proof, those factors remain explicit: for a single frequency with coefficient mass S , the finite tanh construction has bound $2ST_{K,1}/(1 - \rho_K)$. Neglecting the tail correction would convert an absolute budget τ into effective tolerance $\tau/(2S)$. The practical rule assumes this normalization into e_{tol} ; it does not estimate or require S . For arbitrary targets a rough representative frequency also remains a heuristic, so this conversion alone is not a certificate.

To apply either rule, select a candidate interval for λ , evaluate the score at both ends, and solve $\text{score}(\lambda) = e_{\text{tol}}$ by bisection if the values bracket the tolerance. Use the largest feasible endpoint if the full interval lies below budget; if the full interval is above budget, change the interval or the budget. Finally set $\gamma = \lambda/h$. The standalone *choose_lambda.py* implements this calculation in log arithmetic. For tanh with $N = 128$, $L = 2$, $\bar{\omega} = 30\pi/7$ and $e_{\text{tol}} = 2^{-52}$, it returns $\lambda \simeq 0.25553$ and $\gamma \simeq 16.354$. Across the displayed first-pair choices, $\rho_K < 2.29 \times 10^{-7}$, so the omitted geometric correction is below 0.000023%. The plots use $e_{\text{tol}} = 2^{1-p}$ as a reproducible starting choice, with no amplitude input. Their red lines have been recomputed from the simplified first-pair equation; their observed errors are unchanged.

The main text uses a non-strict budget inequality, so a continuous score on a closed admissible interval attains its largest feasible λ whenever the feasible set is nonempty. The supplied routine returns the feasible side of a bisection; the plotted markers show the continuous boundary roots.

The frequency refinement and its limits. One possible estimate is the amplitude-weighted mean $\bar{\omega} = \sum_j |b_j| |\omega_j| / \sum_j |b_j|$. For tanh, the leading small-angle term is linear in $|\omega|$, so this mean reproduces the leading weighted sum when all contributing modes are in that regime. At finite angles this is an approximation; a small high-frequency coefficient can contribute far more than evaluating the bound at the mean suggests. Retain the weighted sum for a certificate. For an interval-only analytic target,

the class estimate (A.11) already applies; a spectral description supplies optional practical information, not a new global Fourier assumption. A real-interval approximation alone does not control the complex values needed to transfer the pole estimate.

The supplied least-squares plots compare these bandwidth predictions with sampled relative errors. They test the usefulness of the rules; their finite fitted networks and error metric are different from the explicit representation and continuous supremum norm bounded here. In particular, the additional general- r identity does not extend the local tanh theorem to GELU, Swish, or Gaussian without a corresponding construction.

Why choose the largest admissible bandwidth? At fixed N , halo design, and target class, the representation theorem gives the non-replica and replica envelopes

$$F(\lambda) = \frac{128\pi B}{\lambda\delta} e^{-W \min\{\pi\delta, \lambda a_H^2\}/8}, \quad G(\lambda) = \frac{128\pi B}{\lambda\delta W} e^{-\pi^2/\lambda}. \quad (\text{A.15})$$

The opening uses $c = \min\{\pi, a_H^2\}/8$ and absorbs the fixed factor $128\pi B/\delta$ into its implicit constant.

For $0 < \lambda \leq 1$, F strictly decreases, while $(\log G)' = (\pi^2 - \lambda)/\lambda^2 > 0$. Provided $(R + 1/2)h \leq \Delta$ and $N \geq 16 \max\{1, a_H^2\}$, admissibility requires $\lambda \in [\lambda_{\min}, 1]$, where $\lambda_{\min} = \max\{8\pi/(\delta N), 2 \log 2/R\}$. Consequently

$$\lambda^* = \max\{\lambda \in [\lambda_{\min}, 1] : G(\lambda) \leq e_{\text{tol}}\}$$

minimizes the proven non-replica envelope over that feasible set. If the set is empty, the width or budget must change. This justifies the budget strategy; it does not identify the minimum of the measured network error. One may use the sharper bound $32\pi B U(\pi^2/\lambda)/(\lambda\delta N)$ instead of G .